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1. Introduction

1.1. Purpose

The purpose of this document is to describe the design of the **hardware-focused components** of the ReMesh system and their supporting embedded software interactions. It serves as a technical design reference for academic evaluation, system reviewers, and developers by documenting architectural decisions, design rationale, constraints, and interfaces related to physical devices. This document does not describe the functional requirements or end-user features of the system; instead, it focuses on *how the hardware-oriented system is structured and designed*.

1.2. Scope

This design document covers the hardware components of the ReMesh system and the embedded software behaviors that directly depend on them. This includes sensor nodes, gateway devices, and end devices built on ESP32 platforms, their communication using LoRa and Wi-Fi, power management considerations, and inter-device data exchange models. Cloud platforms, large-scale backend services, and purely software-based analytics are outside the scope of this document unless they directly influence hardware behavior or embedded firmware design decisions.

1.3. Definitions, Acronyms, and Abbreviations

Term	Description
IoT	Internet of Things
LPWAN	Low Power Wide Area Network
LoRa	Long Range wireless communication
ESP32	Microcontroller with Wi-Fi and Bluetooth
Gateway	Node that bridges LoRa and Wi-Fi
Heartbeat	Periodic gateway broadcast containing latest sensor state
Edge Computing	Local data processing near source

1.4. Overview

ReMesh is a distributed, hardware-centric monitoring system designed for deployment in flood-prone and disaster-affected environments such as riverbanks, dams, and flood plains. The system prioritizes resilience, low power consumption, and autonomous operation under adverse conditions, including network outages and power interruptions. By leveraging low-power sensor devices, long-range wireless communication, and localized data access through gateway nodes, ReMesh enables early flood detection, situational awareness, and emergency information dissemination at the community level.

2. Overall Description

This section provides a stable, high-level description of the ReMesh system with a focus on its hardware-oriented software design. It expands on the overview presented in Section 1.5 by explaining how the system is positioned, how it operates in general terms, and what major elements and constraints shape its design. The descriptions in this section are intentionally general and architectural in nature so that they remain valid across future versions of the system.

ReMesh is designed as a distributed embedded system that combines low-power sensing, long-range wireless communication, and localized data access. The system emphasizes resilience and autonomy, enabling continued operation during network failures, power outages, and extreme environmental conditions. Core system intelligence is distributed across sensor nodes and gateways rather than centralized in cloud infrastructure, reducing single points of failure.

2.1. Product Perspective

ReMesh can be viewed as an alternative to conventional flood monitoring and environmental sensing platforms that rely heavily on continuous internet connectivity and centralized cloud processing. Traditional solutions typically transmit sensor data directly to remote servers, making them vulnerable during disasters when connectivity is degraded or unavailable.

In contrast, ReMesh adopts an edge-centric and mesh-aware design. Sensor nodes communicate using LoRa, a low-power wide-area technology well-suited for long-range and disaster-prone deployments. Gateway devices provide localized data access and coordination, allowing the system to function independently of external infrastructure. This design aligns with emerging trends in edge computing and resilient Internet of Things (IoT) systems.

2.1.1. System Interfaces

The ReMesh software runs primarily on ESP32-based embedded platforms and interfaces with the underlying operating system abstractions provided by the ESP32 runtime environment and real-time operating system (RTOS). These interfaces include task scheduling, timing services, non-volatile storage access, networking stacks for Wi-Fi communication, and hardware abstraction layers for peripheral access. The system avoids complex OS dependencies to ensure portability and reliability on constrained devices.

2.1.2. User Interfaces

User interaction with the ReMesh system is intentionally minimal and hardware-driven. The primary user interface is a lightweight, web-based dashboard hosted directly on gateway devices. This interface is designed around simplicity and clarity, allowing users to view real-time sensor readings, device status, and basic alerts using standard web browsers. The design principles prioritize accessibility, low resource usage, and offline availability rather than rich graphical features.

2.1.3. Hardware Interfaces

The ReMesh software interfaces directly with several hardware components, including the ESP32 microcontroller, LoRa transceiver modules, and a range of environmental sensors such as pH, turbidity, water flow, rain detection and level sensors, temperature and humidity sensors, and GPS modules. Additional interfaces include power management components such as solar charging controllers and battery monitoring circuits. These hardware interfaces strongly influence software design decisions related to timing, power management, and data acquisition.

2.1.4. Software Interfaces

Internally, the ReMesh system is composed of modular firmware components responsible for sensor management, communication handling, data encoding, and local storage. At the gateway level, the firmware interfaces with a web-based frontend served over HTTP. While current versions do not rely on external software systems, the architecture is designed to support future integration with cloud services, analytics platforms, or emergency notification systems without fundamental redesign.

2.1.5. Communication Interfaces

Communication within the ReMesh system is achieved through multiple interfaces. LoRa is used for long-range, low-power communication between sensor nodes, gateways, and end devices. Wi-Fi is used by gateway devices to provide local network access for user devices. Communication protocols are designed to be lightweight and resilient, using compact JSON-based message formats optimized for constrained bandwidth and intermittent connectivity.

2.1.6. Memory Constraints

The ReMesh software is developed with strict awareness of the limited memory resources available on embedded platforms. RAM usage is constrained by the ESP32 architecture, requiring careful management of buffers, task stacks, and data structures. Flash memory usage is optimized by minimizing firmware size and avoiding unnecessary dependencies. Temporary data storage is limited to essential operational data to ensure long-term stability.

2.1.7. Operations

Normal operation of the ReMesh system involves periodic sensor data acquisition, scheduled LoRa transmissions, reception and aggregation of data at gateway devices, and local data visualization. Special operations include initial device provisioning, recovery after power loss, firmware updates, and emergency broadcast modes during disaster events. These operations are designed to require minimal user intervention.

2.1.8. Site Adaptation Requirements

ReMesh deployments are intended for outdoor environments such as riverbanks, dams, and flood plains. Site adaptation requirements include appropriate physical mounting, weather-resistant enclosures, sufficient solar exposure for energy harvesting, and configuration of communication parameters to suit local terrain and distance constraints. The software must tolerate environmental variability without requiring frequent reconfiguration.

2.2. Product Functions

At a functional level, ReMesh performs continuous environmental monitoring, long-range wireless data transmission, local data aggregation, and real-time visualization. The system supports gateway coordination through periodic heartbeat messages and is designed to accommodate future enhancements such as predictive analytics, adaptive sampling, and emergency alert dissemination. These functions are implemented with a focus on reliability and energy efficiency.

2.3. User Characteristics

Typical users of the ReMesh system include disaster management personnel, infrastructure operators, and local authorities with moderate technical proficiency. Community-level users accessing the system through gateway dashboards are expected to have minimal technical knowledge. As a result, the system emphasizes ease of use, clear information presentation, and low operational complexity.

2.4. Constraints

The design of ReMesh is constrained by several factors, including limited bandwidth and payload sizes imposed by LoRa communication, restricted processing and memory resources on embedded hardware, and energy limitations due to reliance on solar power. Environmental conditions such as humidity, temperature extremes, and physical accessibility further constrain hardware and software design choices.

2.5. Assumptions and Dependencies

The system assumes reliable operation of ESP32 hardware platforms and continued availability of LoRa communication within the intended deployment range. It is assumed that deployment sites provide adequate solar exposure to sustain autonomous operation. Dependencies include development toolchains and libraries supporting ESP32 and LoRa modules. Future system extensions may assume intermittent or optional cloud connectivity but do not depend on it for core functionality.

2.6. Apportioning of Requirements

The implementation of ReMesh is expected to follow a phased approach. Initial development prioritizes sensor data acquisition and LoRa communication. Subsequent phases focus on gateway aggregation and visualization, followed by gateway synchronization and resilience improvements. Advanced features such as analytics, prediction, and extended alerting mechanisms are planned for later stages.

3. Specific Requirements

This chapter defines the detailed technical and architectural requirements of the ReMesh system with a focus on hardware-oriented embedded software behavior. The requirements described here are intended to guide implementation while remaining independent of specific firmware versions or future feature expansions.

3.1. External Interface Requirements

3.1.1. User Interfaces

User interaction with the ReMesh system is primarily provided through a **local web-based interface** hosted on gateway devices. This interface is accessed via standard web browsers on mobile phones or laptops connected to the gateway's Wi-Fi network. The interface presents real-time sensor readings, device identification details, gateway status, and basic system health indicators such as signal strength and last update time. The design prioritizes clarity, minimalism, and readability under emergency conditions, ensuring that non-technical users can quickly understand the displayed information. Configuration options are limited to prevent accidental misconfiguration and are intended for authorized technical personnel only.

3.1.2. Hardware Interfaces

The ReMesh software interfaces with multiple hardware components across its device types. Sensor nodes interface with environmental sensors including pH, turbidity, water flow, rain detection and rain level sensors, temperature and humidity sensors, and a GPS module using analog, digital, and serial communication protocols. LoRa transceiver modules interface with the ESP32 microcontroller via SPI. Gateway and end devices additionally interface with Wi-Fi radios integrated into the ESP32 platform. Power management interfaces include battery voltage monitoring and solar charging controllers to support autonomous operation.

3.1.3. Software Interfaces

Internally, the ReMesh firmware is organized into modular components responsible for sensor acquisition, packet formatting, communication scheduling, power management, and local data presentation. At the gateway level, the firmware interfaces with an embedded web server and a lightweight frontend application for visualization. External software interfaces are intentionally minimal, allowing future integration with cloud dashboards, analytics platforms, or machine learning services without impacting core embedded functionality.

3.1.4. Communication Interfaces

Communication within the ReMesh system relies on LoRa for long-range, low-power data exchange between sensor nodes, gateways, end devices, and neighboring gateways. Data packets are encoded in compact JSON structures optimized for bandwidth efficiency. Wi-Fi communication is used by gateway devices to provide local access to users through HTTP-based web services. Optional future internet connectivity may be used for data synchronization with centralized dashboards but is not required for core system operation.

3.2. Architectural Design (Embedded Systems)

3.2.1. High-Level Architecture

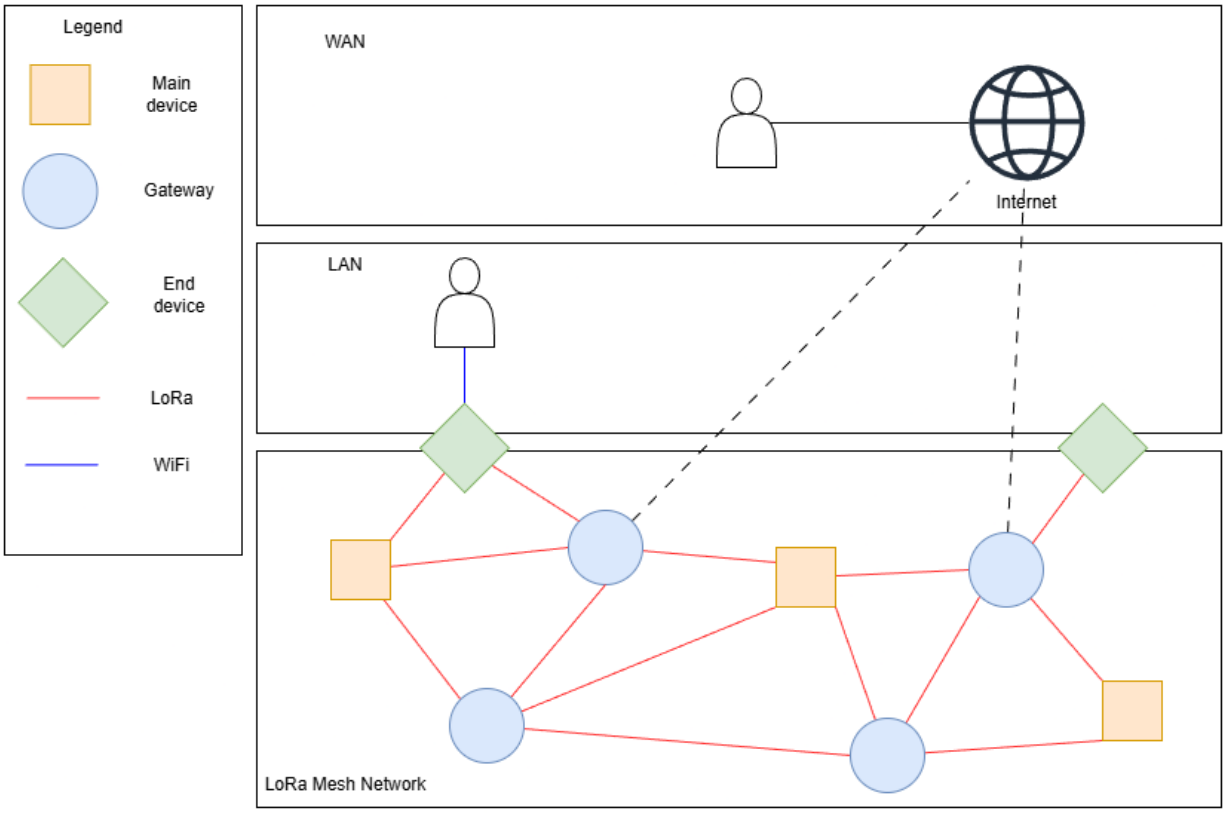

Figure 1: High-Level Archtecture of the system

ReMesh adopts a distributed, edge-centric architecture consisting of three primary device roles: sensor nodes, gateway devices, and end devices. Sensor nodes collect and broadcast environmental data via LoRa. Gateway devices receive this data, visualize it locally, and rebroadcast summarized heartbeat messages containing the most recent sensor state. End devices receive both direct sensor transmissions and gateway heartbeats for redundancy and future analytical processing. The architecture minimizes reliance on centralized infrastructure and supports resilient operation during network disruptions.

3.2.2. Hardware and Software Requirements with Justification

The ESP32 microcontroller is selected due to its low power consumption, integrated Wi-Fi capabilities, and strong community and tooling support. LoRa technology is chosen for its long-range communication and suitability for low-bandwidth disaster monitoring scenarios. The combination enables cost-effective deployment while meeting the requirements for range, power efficiency, and reliability. The use of solar panels and battery packs supports sustained autonomous operation in remote locations.

3.2.3. Risk Mitigation Plan and Alternative Solutions

Key risks include LoRa packet collisions, power depletion during prolonged low-sunlight conditions, and hardware degradation due to environmental exposure. These risks are mitigated through controlled transmission intervals, power-aware scheduling, redundant gateway deployment, and weather-resistant enclosures. Alternative solutions such as multi-channel LoRa configurations, external power sources, or higher-capacity batteries are considered for critical deployment sites.

3.2.4. Cost Benefits Analysis

The proposed solution uses low-cost, commercially available hardware components while delivering significant societal benefits through early flood detection and situational awareness. By reducing dependence on continuous internet connectivity and centralized infrastructure, ReMesh lowers operational costs and increases deployment feasibility in resource-constrained regions. The cost-benefit balance supports scalability across multiple monitoring locations.

3.3. Performance Requirements

The system is required to transmit sensor data within acceptable latency bounds defined by LoRa communication constraints, typically on the order of seconds per transmission. Gateway devices must respond to local user requests within a few seconds over Wi-Fi. Firmware execution must remain within the memory and processing limits of the ESP32 platform while maintaining stable long-term operation.

3.4. Design Constraints

Design constraints include limited LoRa bandwidth, strict power consumption requirements due to solar operation, and constrained memory and processing resources on embedded devices. These constraints influence communication protocols, packet sizes, firmware architecture, and data processing strategies.

3.5. Software System Attributes

3.5.1. Reliability

ReMesh is designed for continuous operation in harsh environmental conditions. Redundant data reception paths and conservative power management strategies improve reliability and reduce data loss.

3.5.2. Availability

Local gateways ensure system availability even during internet outages. The decentralized architecture allows the system to continue functioning during partial network failures.

3.5.3. Security

Basic security measures include device identification, controlled access to configuration interfaces, and restricted Wi-Fi connectivity. Security mechanisms are designed to balance protection with low system complexity.

3.5.4. Maintainability

The modular firmware design supports ease of maintenance and future enhancements. Components likely to evolve include communication protocols, analytics modules, and power optimization logic.

3.6. Other Requirements

The system shall support future extensions including machine learning-based flood prediction, emergency text or voice message dissemination, and enhanced energy optimization techniques.

4. Appendices

Appendix A: Problem Background

Sri Lanka has experienced recurring flood events resulting in significant loss of life, infrastructure damage, and economic disruption. Many affected regions lack real-time monitoring infrastructure or depend on centralized systems that fail during extreme weather or network outages. ReMesh is designed in response to these challenges, focusing on decentralized monitoring, local data accessibility, and resilient communication to support early warning and disaster response efforts.

Appendix B: Sample Data Packet Formats

The following example illustrates a typical sensor data packet transmitted by a ReMesh sensor node:

```
{
 "device_id": "device_001",
 "gps": {
 "lat": 0.000000,
 "lng": 0.000000,
 "speed_kmph": 0.00,
 "sats": 0
},
 "temp_c": 30.00,
 "hum": 56.00,
 "flow_lpm": 1.000,
 "rain1_pct": 0,
 "rain2_pct": 0,
 "turbidity_pct": 0.0,
 "ph": 7.00,
 "ts": 60
}
```

Gateway heartbeat packets encapsulate the most recent sensor data along with gateway identification:

```
{
 "gateway_id": "gateway_001",
 "last_rx": { /* latest sensor packet */ }
}
```

Appendix C: Deployment Considerations

ReMesh devices are intended for outdoor deployment near rivers, dams, and flood plains. Proper installation requires weather-resistant enclosures, stable mounting structures, and sufficient solar exposure for energy harvesting. Antenna placement and orientation are critical for reliable LoRa communication.

Appendix D: Packaging and Media Instructions

Firmware images, configuration files, and documentation are packaged digitally for deployment and evaluation. Devices are labeled with unique identifiers corresponding to their logical roles within the ReMesh network. Demonstration media, including videos and screenshots of the gateway web interface, may be provided alongside the code repository for academic assessment.

Appendix E: User Feedback Summary

Initial prototype demonstrations indicate that users value the system's ability to provide local access to flood-related data without reliance on internet connectivity. Feedback emphasizes the importance of clear visual indicators, simple status information, and reliable operation during adverse conditions. This feedback informs future refinements of the interface and system behavior.

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